The rank-four toroidal families in cubic coordinates

Human Compression project

The fixed manuscript [1] displays three period families for each of the types $\widetilde{B}_3$ and $\widetilde{A}_3$ on page 11. They follow directly from the acquired cubic halving construction. We give the exact change of parameters, including the smallest allowed periods. This proves existence and the marked group descriptions of the displayed families. Their exhaustion is the classification attributed in the source to Ens; it is not a consequence of this parameter calculation.

1 The acquired construction

Put

$$D = \{v \in \mathbb{Z}^3 : v_1 + v_2 + v_3 \equiv 0 \pmod 2\}, \qquad E = 2\mathbb{Z}^3 + \mathbb{Z}\mathbf{1}.$$

Write B_3 for the signed permutation group, and B_3^+ for the subgroup with an even number of sign changes. Let z_i change the sign of coordinate i, let s_i interchange coordinates $i, i+1$, and let τ_v be translation by v. The acquired affine calculation gives

$$x = \tau_{(1,1,0)} z_1 z_2 s_1, \qquad y = z_2 z_3 s_2,$$

$$\langle x, s_1, s_2, z_3 \rangle = D \rtimes B_3, \qquad \langle x, s_1, s_2, y \rangle = D \rtimes B_3^+.$$

Their diagrams have, respectively, the bonds $02, 12$ labelled three and 23 labelled four, and the four-cycle $02, 12, 13, 03$ with all labels three. Thus these are the marked affine groups of types $\widetilde{B}_3$ and $\widetilde{A}_3$.

For $t \geq 2$ put

$$\Lambda_1(t) = t\mathbb{Z}^3, \qquad \Lambda_2(t) = tD, \qquad \Lambda_3(t) = tE, \qquad L_k(t) = D \cap \Lambda_k(t).$$

The acquired geometry proof applies one halving at the left end of the cubic toroid, and then one at the opposite end. It gives actual regular hypertopes H_{10} and H_{11}, with the above markings and groups

$$(D/L_k(t)) \rtimes B_3, \qquad (D/L_k(t)) \rtimes B_3^+,$$

for every $k \in \{1, 2, 3\}$ and $t \geq 2$ except $(k, t) = (1, 2)$. Here existence includes thinness, residual connectivity and chamber transitivity, proved in the acquired incidence models. These conclusions are not being inferred from the diagrams or group orders. In particular, the case $(k, t) = (2, 2)$ is allowed for H_{10} and H_{11}: its separate exclusion concerns a second halving at the same end.

2 The source parameters

The period generated by a source vector means the integer span of its orbit under the corresponding finite linear group. Both finite groups above give the same three spans in dimension three.

Proposition 1. *The following table identifies every displayed source family with the acquired construction. The orders in the last two columns are the orders of the full type-preserving groups.*

source vector and domain	(k,t)	$L_k(t)$	$\lvert G_{\widetilde{B}_3}\rvert$	$\lvert G_{\widetilde{A}_3}\rvert$
$(2s,0,0),\ s\geq 2$	$(1,2s)$	$2s\mathbb{Z}^3$	$192s^3$	$96s^3$
$(s,s,0),\ s\geq 2$	$(2,s)$	sD	$48s^3$	$24s^3$
$(2s,2s,2s),\ s\geq 1$	$(3,2s)$	$2sE$	$768s^3$	$384s^3$

The source objects are the corresponding incidence quotients of the universal marked affine complexes.

Proof. The orbit of e_1 under either group is the set of signed coordinate vectors: an unwanted sign change can be placed in a zero coordinate. Its span is $\mathbb{Z}^3$. Similarly the orbit of e_1+e_2 consists of all roots $\pm e_i \pm e_j$. These span D: for example $e_1+e_2, e_2-e_1, e_3-e_2$ are a lattice basis of determinant two.

Under B_3, the orbit of $\mathbf{1}$ spans E, since sign differences give $2e_i$. Under B_3^+ it spans $M = 2D+\mathbb{Z}\mathbf{1}$: differences from $\mathbf{1}$ are twice the characteristic vectors of two-element subsets, whose span is D. In dimension three $M=E$. Indeed $\mathbf{1}-e_i \in D$, and $2e_i = 2\mathbf{1}-2(\mathbf{1}-e_i) \in M$. The reverse inclusion is immediate. Scaling gives precisely the three source period lattices in the table.

Each of these lattices lies in D. The first and third do so because their coordinates are even, and the middle does by definition. Consequently their intersections with D are unchanged. Substitution in $L_k(t)$ proves the parameter correspondence. The first row has $t\geq 4$, the second has $t\geq 2$ with $k=2$, and the third has $t\geq 2$ with $k=3$. None is the excluded pair $(1,2)$, so both acquired geometry constructions apply throughout the exact source domains, including $s=1$ in the third row.

Finally $[\mathbb{Z}^3 : D] = 2$, $[\mathbb{Z}^3 : E] = 4$, and scaling a lattice by a multiplies its index by a^3. Thus the three indices in D are $4s^3, s^3, 16s^3$. Multiply these by $|B_3| = 48$ or $|B_3^+| = 24$. The acquired coset identification of each actual geometry gives the stated incidence quotient, with its actual marking. □

Attribution and scope. The source attributes the complete rank-four family classification to Ens. The cubic toroid input is the retained McMullen–Schulte classification, and the halving constructions originate with the earlier authors cited in the acquired proof. The present addition is the literal parameter and marking interface between those proofs and this source. It does not prove that every invariant lattice is in the table, relax a small-period condition, or claim a new conjecture answer. No new finite census, Lean formalization or compression measurement is involved.

References

[1] M. E. Fernandes, D. Leemans and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*, fixed catalogue L033 manuscript, pages 10–11.

[2] Human Compression project, *Generalized halving: corrected source recovery*, accepted affine group calculation and iterated geometry proof, Sections 20 and 30 in the frozen application sources.